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TECHNICAL SUMMARY

Power Factor Correction

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ABSTRACT

ePLL-based power factor correction control with a regulated 400 V DC output, validated in real-time HIL. This project develops a PFC control strategy in MATLAB/Simulink. An enhanced phase-locked loop (ePLL) provides accurate grid synchronization, an outer loop regulates the DC output at 400 V, and an inner loop shapes the input current to raise the power factor and reduce harmonics. It was validated in real time on PLECS RT Box.

Regulated DC output	400 V
Grid synchronization	ePLL
Power factor	0.99
Input current THD	3.2%

1. Problem and motivation

A plain diode rectifier draws current in short, distorted pulses. That gives a poor power factor and high harmonic content, wastes distribution capacity and can break grid harmonic limits such as IEC 61000-3-2.

A PFC stage shapes the input current to follow the grid voltage, but only if its control stays accurately synchronized to the grid and keeps the DC output stable as conditions change.

2. Methodology

- Developed the PFC control strategy in MATLAB/Simulink with an ePLL for accurate grid synchronization
- Designed the control architecture to regulate the DC output at 400 V while improving power factor and reducing input current harmonics
- Performed real-time HIL validation on PLECS RT Box for rapid prototyping and verification
- Tested the control under dynamic operating conditions

3. Figures

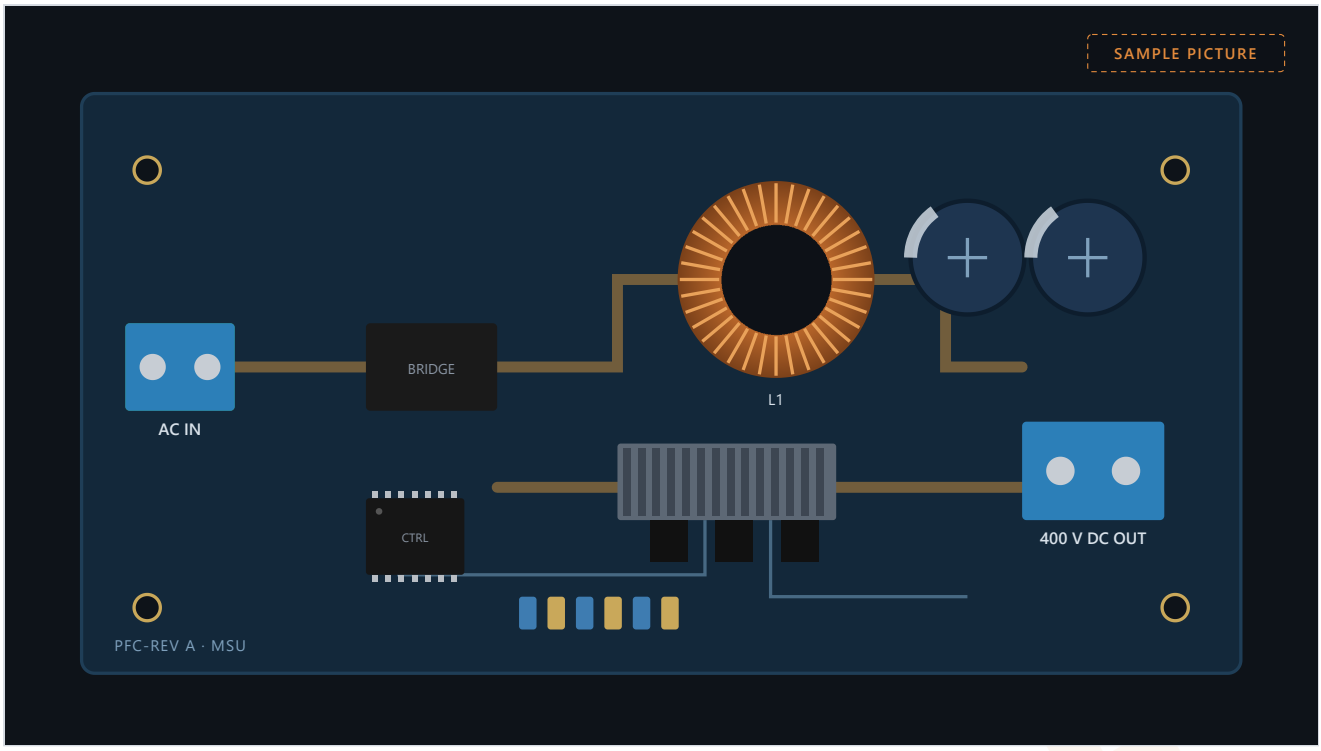


Figure 1. PFC board: AC input, bridge, boost inductor, switches on the heatsink, and the 400 V DC output.

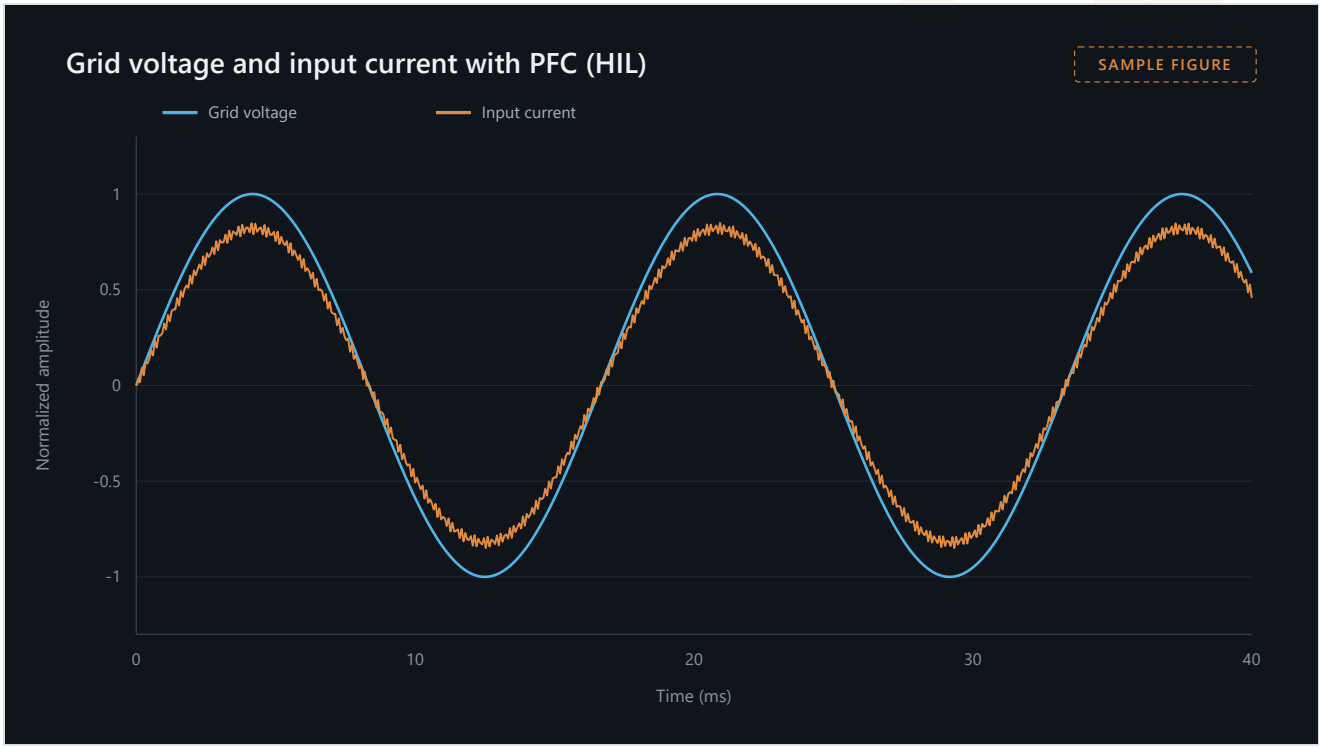


Figure 2. Grid voltage and input current in phase, from HIL.

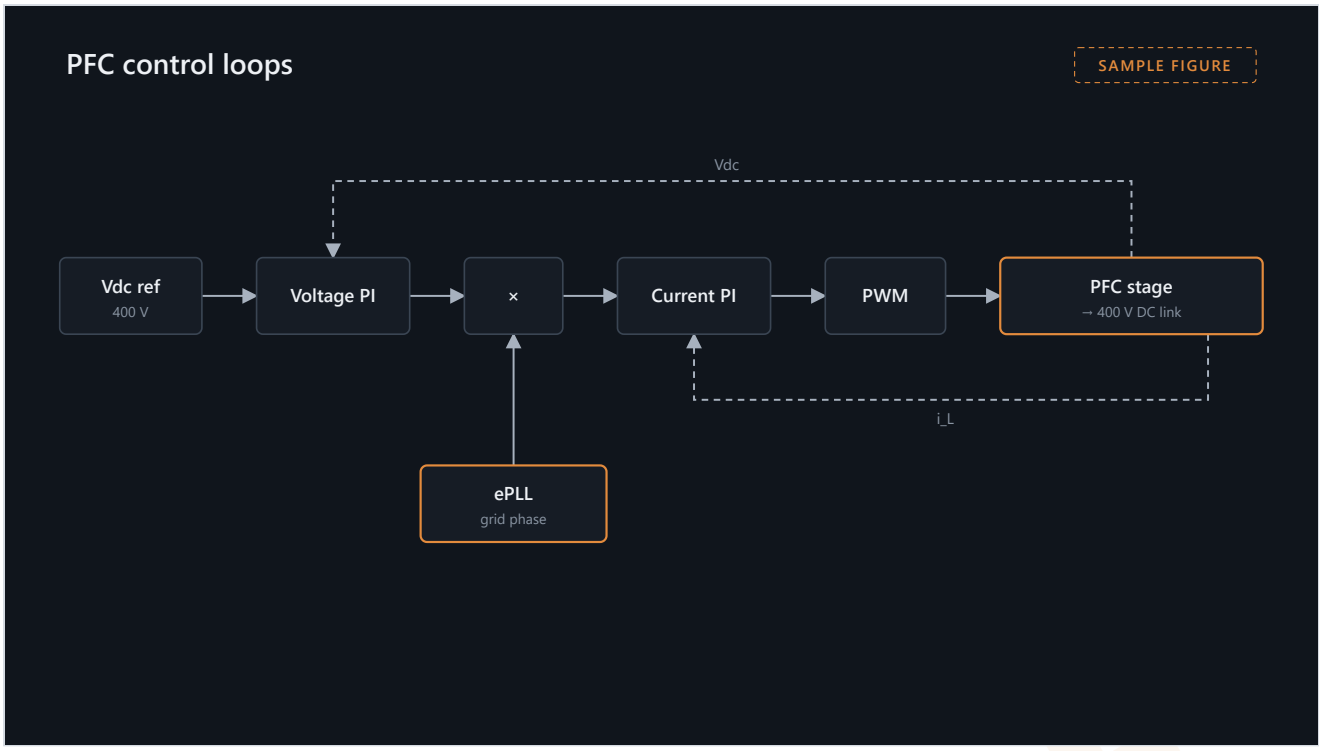


Figure 3. Control structure: voltage loop, ePLL and current loop.

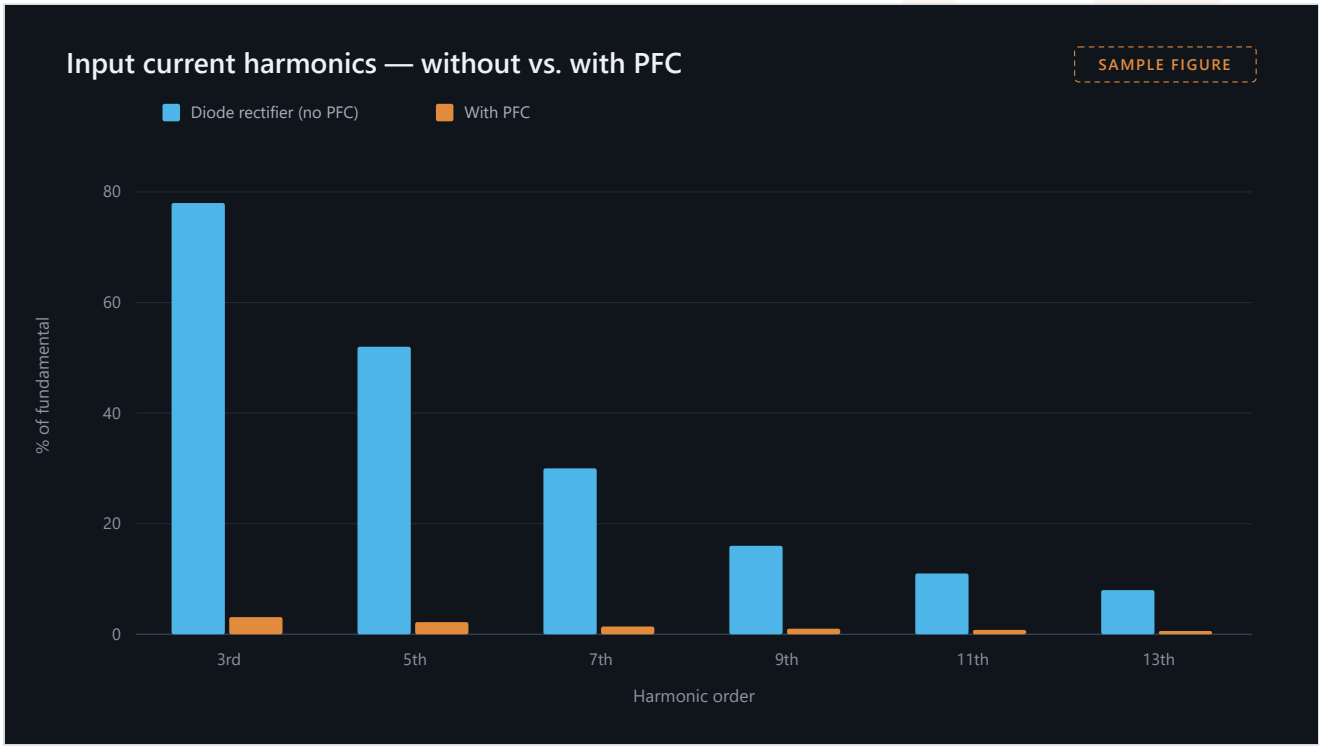


Figure 4. Input current harmonics with and without PFC.

4. Results

- DC output regulated at 400 V
- Power factor of 0.99 and input current THD of 3.2% in HIL

5. Team and contribution

Contribution: Developed the PFC control strategy and the ePLL, designed the control architecture, and ran the real-time HIL validation.

Tools: MATLAB/Simulink, PLECS, RT Box, ePLL, PI control

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